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Results of ruthenium irradiation of uveal melanoma

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Abstract

The follow-up results are presented for the radiotherapeutic treatment of 49 patients with uveal eye melanoma using ruthenium-106 ($^{106}Ru/^{106}Rh$) applicators. Graded doses were applied and the degree of regression was found to be correlated with the dose at the top of the tumour. Complete regression is found in 66% of the patients treated with a top dose above 150 Gy, in which case the initial tumour prominence diminishes in about three years to a stable disease or to a flat scar. At this dose a marginal increase in complications was observed as compared with lower dose groups. Useful vision could be preserved in 75% of the eyes.

Key words: Radiotherapy; Brachytherapy; Ophthalmology; Melanoma; Ruthenium application

1. Introduction

Until a few decades ago the only curative therapy for uveal melanoma was thought to be enucleation of the eye. Enucleation, however, is a mutilating treatment and therefore radiotherapy with the possibility of preserving the eye has been explored as an alternative. Stallard [10] was the first to use ^{60}Co plaques for irradiation of the eye with reasonable success. The disadvantage of ^{60}Co, however, is the deep penetration of the radiation, leading to an unnecessary irradiation of the healthy tissue. Therefore, ruthenium ($^{106}Ru/^{106}Rh$) applicators were developed and used with promising results by Lommatzsch and coworkers [4–8].

Following the good results of Lommatzsch, in May 1984 we started the treatment of uveal melanoma with radiation therapy using ruthenium applicators with special attention for dose and follow-up. Until January 1991 we irradiated 49 patients with a total of 52 applications. Although an eye conserving treatment has cosmetic advantages, the major potential disadvantage is that a risk of local recurrence remains, in which case it can be a source of distant metastases. If this recurrence can be detected at an early stage, an additional treatment (laser/xenon coagulation, second irradiation or enucleation) is possible. It is, however, more important to avoid a local recurrence as a source for further seeding. Therefore, it is necessary to know which dose can be considered to be tumouricidal and to what size eye melanomas can be treated with ruthenium applicators. As far as the dose is concerned it is not yet known whether the dose in the top or at the base of the tumour is most important. For this reason we analyzed our results with different dose levels with special attention to a possible correlation between regression and dose.

2. Material and methods

Between May 1984 and January 1991, 49 patients with uveal melanoma were treated with ruthenium applicators. All patients were informed about the two possible treatments: enucleation or irradiation. Mostly, patients of 60 years or older were advised to accept irradiation. Younger patients were preferably treated with enucleation unless they had a decreased visual acuity of the other eye or accompanying eye diseases. Irradiation was always performed when enucleation of the eye was

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